CSCI 5526

Understanding Clinical Care and Clinical Data

[Content inspired from MIT Machine Learning for Healthcare class lectures 2&3]

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Announcements

- Final project survey: https://z.umn.edu/ml4h_project_survey (due 9/18)
- Shared sample project topics and datasets.
- MP1 was released (due 9/25).

Student Questions

I read about an article once on EPIC's failure on creating an AI model that predicted a no-show patient. The authors evaluated the AI model and found that the model's training data accounted for the race or the individuals immigration status. I think it is also important to mention that a model just does not fail because of accuracy but it can also fail because of how biased that model is.

-- You are absolutely right. The "No-Show" model was designed to predict which patients would skip appointments so clinics could overbook slots or target those patients with reminders.

-- However, the model relied heavily on demographic data points, including race, ethnicity, socioeconomic markers, and even immigration or documentation status.

-- This is an instance of algorithmic bias.

Student Questions

What validations and tests would be performed in order to be approved by the FDA?

Technical & Analytical Validation -- developers must prove the underlying machine learning model is stable, reproducible, and precise.

Clinical Validation & Utility Testing -- AI system functions accurately and safely when interacting with real-world clinical data.

Human Factors & Usability Engineering

Software Assurance, Lifecycle, & Cybersecurity -- AI systems must meet rigid digital infrastructure security requirements to protect patient data and system integrity.

I know LLM systems like OpenEvidence are popular among physicians. Do these models require FDA approval?

Under current guidance and the Cures Act, software that merely surfaces medical literature, provides reference information, or acts as general educational literature retrieval does not always require active FDA pre-market review.

Student Questions

When it comes to some diseases that have a broad range of diagnostic measures (such as sepsis potentially being caused by many things), does that make it harder or easier to narrow the scope when building an algorithm to help a physician diagnose an individual?

-- Conditions like sepsis are extremely heterogeneous with multiple causes. So building a generalizable algorithm is much harder.

Student Questions

How do you determine the scope of an approaches generalizability? If I train with ECG data from one hospital, should I anticipate it to work in nearby other? Far away others all? Just trial and error?

-- Consider different types of distribution shifts.

-- [Internal Validation] ⟶ [Temporal Validation] ⟶ [External Validation] ⟶ [Multi-Site Pooling]

What drives some datasets to be more easily standardized and shared, such as vaccine records, whereas other data is still more siloed?

-- the diversity of the data structure, commercial and political incentives, technical standardization frameworks, and public health mandates.

Student Questions

Would it better to develop individual inhouse models for each region to overcome the problem of deidentifying people?

-- Small data and rare event problem

-- Local bias and overfitting

-- Massive redundant workload

Would localized, in-house AI models be more sustainable for regional hospitals than universal models from third-party vendors, given the strict privacy and data anonymization requirements?

-- Data is the main limitation.

Student Comments

The slides mentioned rare diseases having small sample sizes and suggested learning from healthy subjects. What is best practice to prevent a model from being overfit to a predominantly healthy population for a rare disease?

-- Convert to unsupervised anomaly detection problem.

-- Leverage self-supervised learning.

Is the avoidance of false positive of a system give extra complexity and workload and how to mitigate it?

-- Yes. Massive engineering burden and clinical burden. Alarm fatigue vs automation complacency.

Student Questions

Many of the deployment failures discussed today were identified by comparing predictions with ground-truth labels, for example through changes in AUC. However, you also mentioned that many healthcare problems are unsupervised, such as discovering disease subtypes. If there is no ground-truth label, how can we detect whether an unsupervised model has failed after deployment or whether the data distribution has changed in a clinically meaningful way? Is comparing the distribution of X or learned representations enough, or do we still need external outcomes or expert validation?

-- We can have measures like reconstruction error, etc to monitor unsupervised methods.

-- But we still need expert validation.

We focused on deployed models that tried to help clinicians and nurses make decisions regarding diagnosis and treatment planning. Have prognosis models faced similar issues, and seen comparable successes?

-- Yes, they have faced the exact same structural failures and bias traps as diagnostic models.

Student Questions

For which job positions in the labor market might this course be useful or required?

-- Any applied AI position. Specifically, in health sector.

Will we work on some specific technology or are there any partnership projects with some medical companies?

-- No partnership projects, unfortunately.

Would it be possible to see a few sample final projects from previous years to understand the expectations and requirements for the final project.

-- Posted!

Outline

- Goals of Health Care
  - Understanding mortality, disability, morbidity
- Tasks of Health Care
  - Diagnosis
  - Prognosis
  - Treatment
  - Prevention/Public Health
- Understanding Clinical Data

What is Health

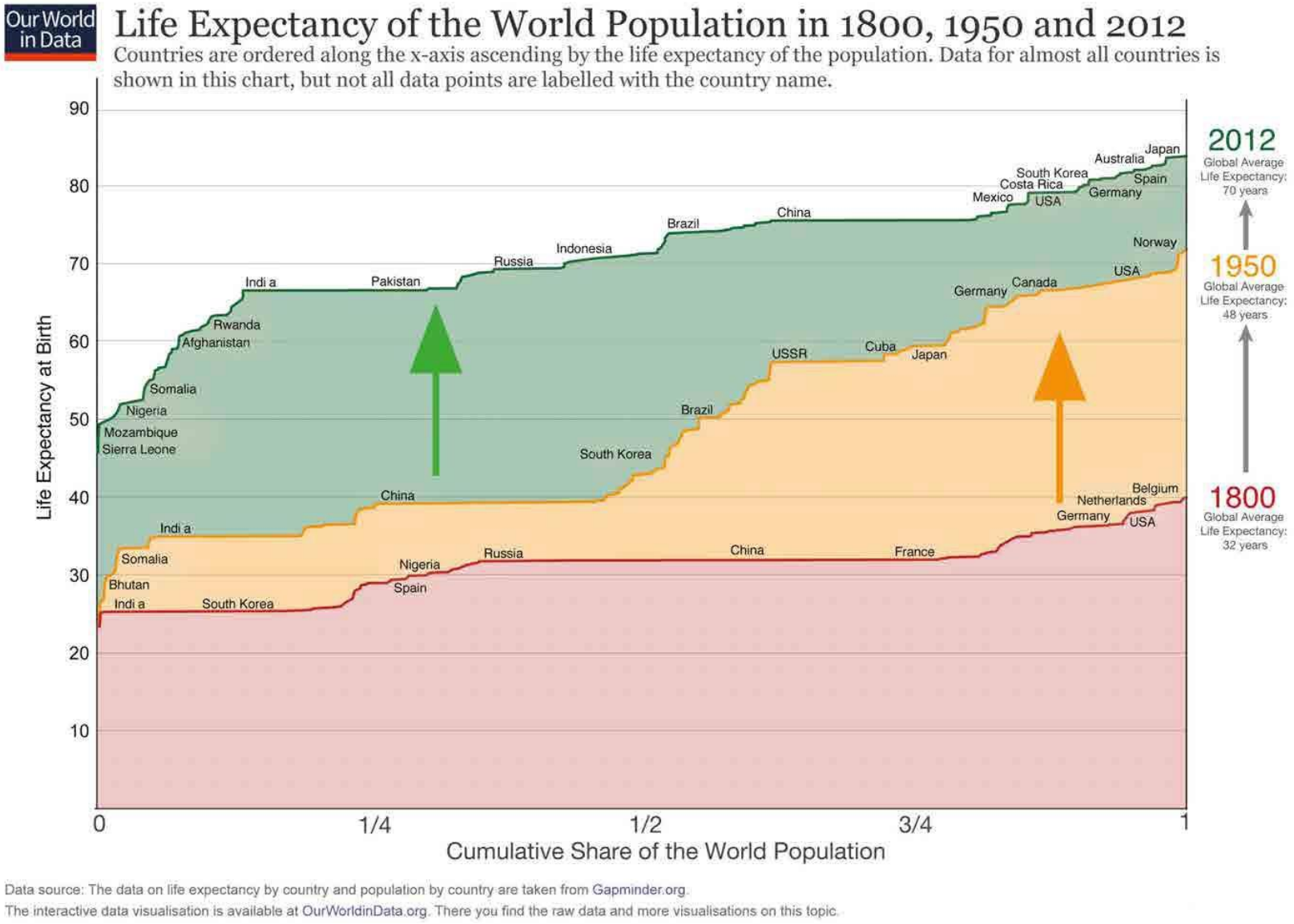
"A state of complete physical, mental and social well-being and not merely the absence of disease or infirmity"

- Physical
- Mental
- Social

  -- very hard to measure

Goals of Healthcare

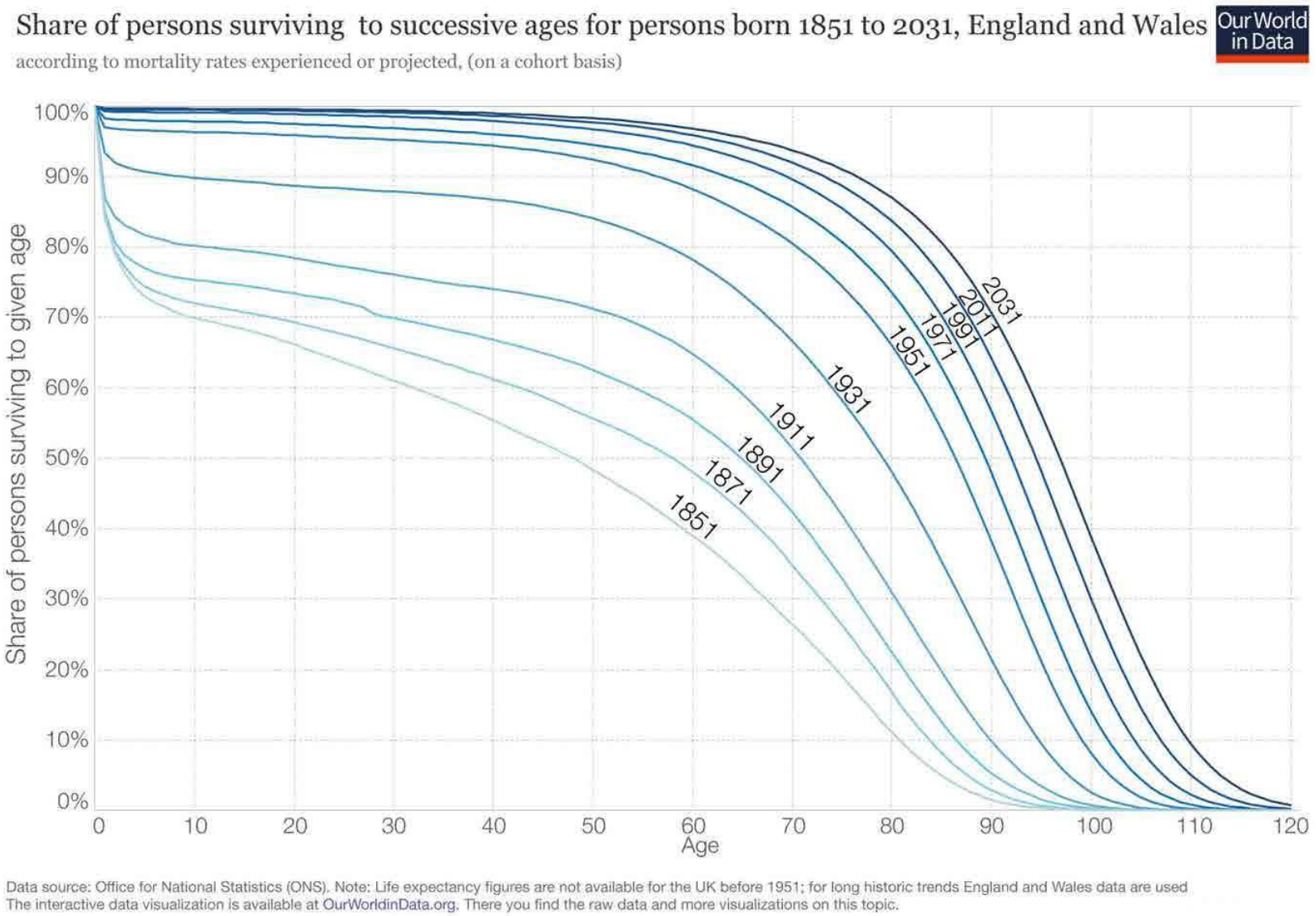
- Reduce
  - Mortality
  - Disability
  - Morbidity



Courtesy of Max Roser and Our World in Data. Used under CC BY.

Longevity at birth (CIA World Fact Book, 2001, 2018)

Country	*Male*		*Female*	
	2018	*2001*	*2018*	*2001*
Rwanda	62.6	38.35	66.5	39.65
South Africa	62.7	47.64	65.6	48.56
Kenya	63.1	46.57	66.1	48.44
Cambodia	62.7	54.62	67.9	59.12
Russia	65.6	62.12	77.3	72.83
Brazil	70.7	58.96	78.0	67.73
Turkey	72.9	68.89	77.7	73.71
Albania	76.0	69.01	81.6	74.87
Israel	80.8	76.69	84.7	80.84
USA	77.8	74.37	82.3	80.05
France	78.9	75.01	85.3	83.01
Japan	82.2	77.62	89.0	84.15



Courtesy of Max Roser and Our World in Data. Used under CC BY.

Causes of death

(USA, 2014)

Cause	Deaths/100K	%
Heart disease	192.7	23.4
	185.6	22.5
ase	46.1	5.6
	42.7	5.2
	41.7	5.1
	29.3	3.6
	24.0	2.9
	17.3	2.1
	15.1	1.8
	13.4	1.6
OTHER	*215.8*	*26.2*
TOTAL	823.7	100.0

Number of deaths for leading causes of death

- Heart disease: 680,981
- Cancer: 613,352
- Accidents (unintentional injuries): 222,698
- Stroke (cerebrovascular diseases): 162,639
- Chronic lower respiratory diseases: 145,357
- Alzheimer's disease: 114,034
- Diabetes: 95,190
- Nephritis, nephrotic syndrome, and nephrosis: 55,253
- Chronic liver disease and cirrhosis: 52,222
- COVID-19: 49,932

2024

https://www.medicalnewstoday.com/articles/282929.php

Morbidity: Top 10 Chronic Conditions

Percentage of Older Adults with Chronic Conditions	
High cholesterol	58.2
Hypertension	56.7
Arthritis	48.7
Cancer	23.1
Diabetes	20.5
Heart disease	17.9
Ulcers	11.3
Stroke	7.2
Asthma	6.9
Kidney disease	5.1
Chronic bronchitis	5.0
Emphysema	4.0

https://courses.lumenlearning.com/suny-lifespandevelopment/chapter/chronic-conditions/

Time Scales in Medicine

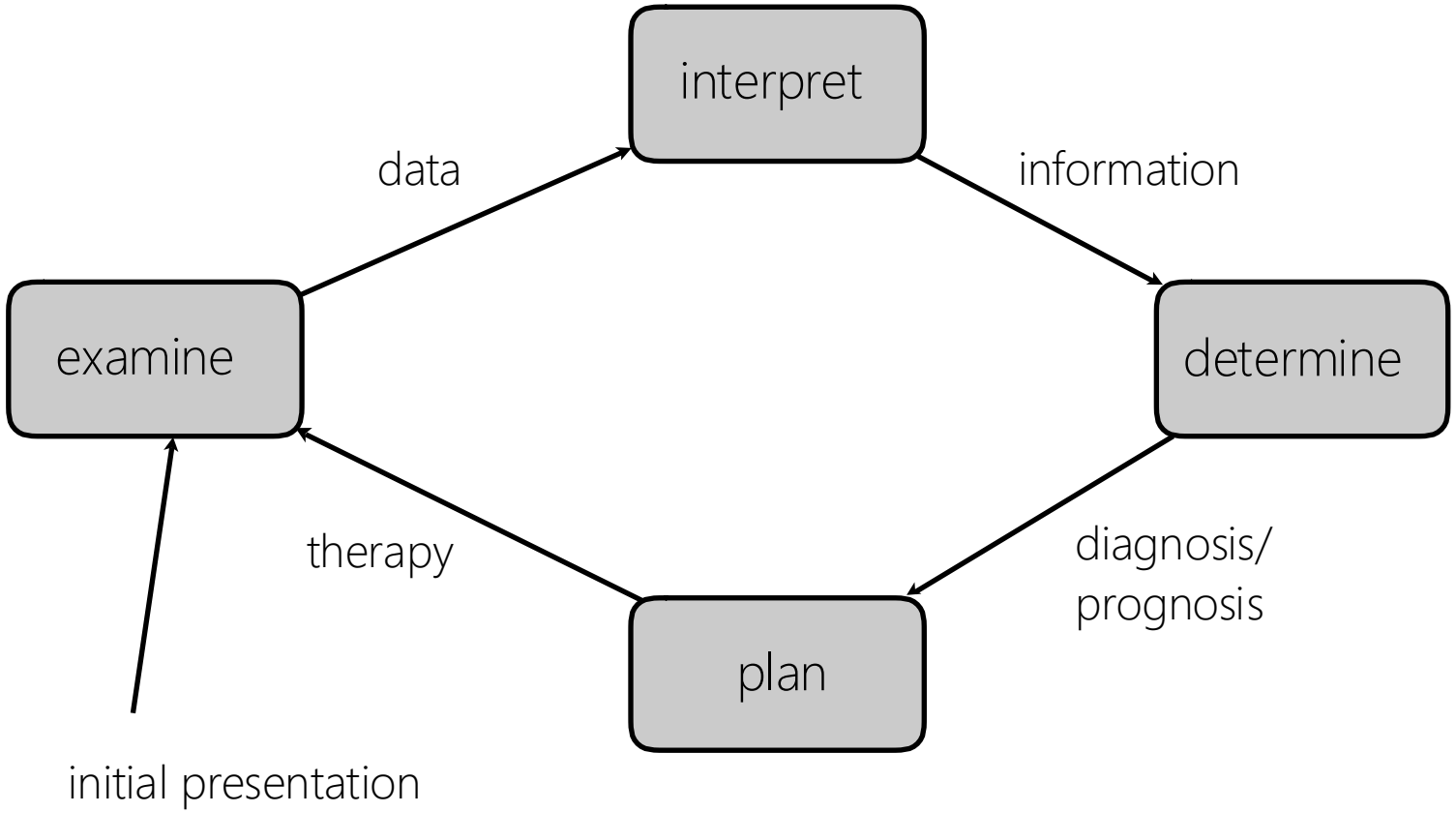
- Cure - usually acute illness
- Manage - long-term, chronic illness
- Prevent
- Predict

Traditional tasks of medicine

- Diagnosis
  - "the art or act of identifying a disease from its signs and symptoms"
- Prognosis
  - "the prospect of recovery as anticipated from the usual course of disease or peculiarities of the case"
- Therapy
  - "therapeutic medical treatment of impairment, injury, disease, or disorder"

Definitions from Meriiam-Webster

The Medical Cycle



Cognitive Theory of Diagnosis

- From initial complaints, guess suitable hypothesis
- Use current active hypotheses to guide questioning
- Failure to satisfy expectations is the strongest clue to a better hypothesis; differential diagnosis
- Hypotheses are activated, de-activated, confirmed or rejected based on
  - logical criteria
  - probabilities based on:
    - findings local to hypothesis
    - causal relations to other hypotheses (coherence)

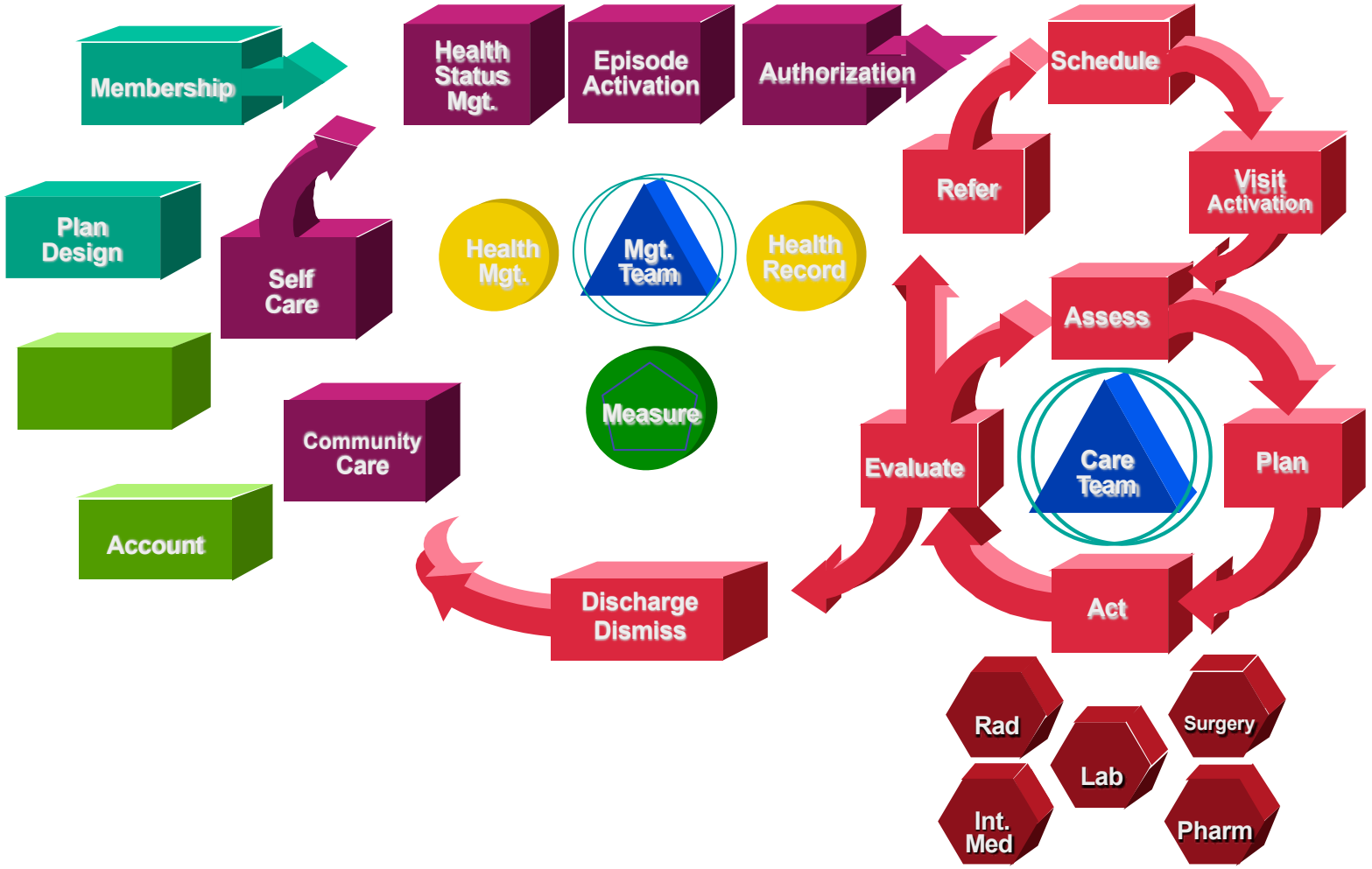
≈ Scientific Method

Pauker, S. G., Gorry, G. A., Kassirer, J. P., & Schwartz, W. B. (1976). Towards the simulation of clinical cognition. Taking a present illness by computer. The American Journal of Medicine, 60(7), 981–996.

Meta-level Processes

- Acquisition and application of knowledge
- Education
- Quality control and process improvement
- Cost containment
- Reference (library)

Enterprise-level Clinical Process Automation



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How Does the Health System Learn?

- "Evidence-based medicine"
  - Contrast with "Tradition-based medicine" — Apprenticeship
- Randomized Controlled Clinical Trial (RCT)
  - E.g., is drug A more effective than drug B for condition X?
  - Narrow selection of patient cases and controls
  - Careful collection of systematically organized data
  - Statistical analysis of outcomes

  => Statistically significant conclusions
- But:
  - **Heterogeneity**: most cases to which RCT results are applied do not fit trial criteria.
  - **Short follow-Up**: trials run for limited times, but use is longer.
  - **Small samples**: some effects are rare but devastating.

The Learning Healthcare System

"One in which progress in science, informatics, and care culture align to generate new knowledge as an ongoing, natural by-product of the care experience, and seamlessly refine and deliver best practices for continuous improvement in health and health care"

- Needs:
  - Comprehensive collation of all clinical, social, demographic, behavioral data
  - Routine capture of novel data sources:
    - genomes, gene expression, etc.
    - environmental factors (e.g., metagenomics)
    - physiological response to life situations
  - Technical infrastructure
    - Storage and analysis of truly "big data"
  - Incentives and demonstrations of utility

- Check out the Center for Learning Health System Sciences at UMN
  - https://med.umn.edu/clhss

Preventive Medicine

- Public Health
  - Tracking disease prevalence
  - Tracing infections
  - Quarantine

Tracking Disease Prevalence by Systematic Classification

- 17th century: John Graunt on the London Bills of Mortality estimated mortality before age 6 at 36%
- 18th century: Sauvage, Linnaeus, Cullen made first attempts at systemic classification
- 1853—first International Statistical Congress led to Wm. Farr's system:
  - epidemic diseases
  - constitutional (general) diseases
  - local diseases arranged according to anatomical site
  - developmental diseases
  - diseases that are the direct result of violence

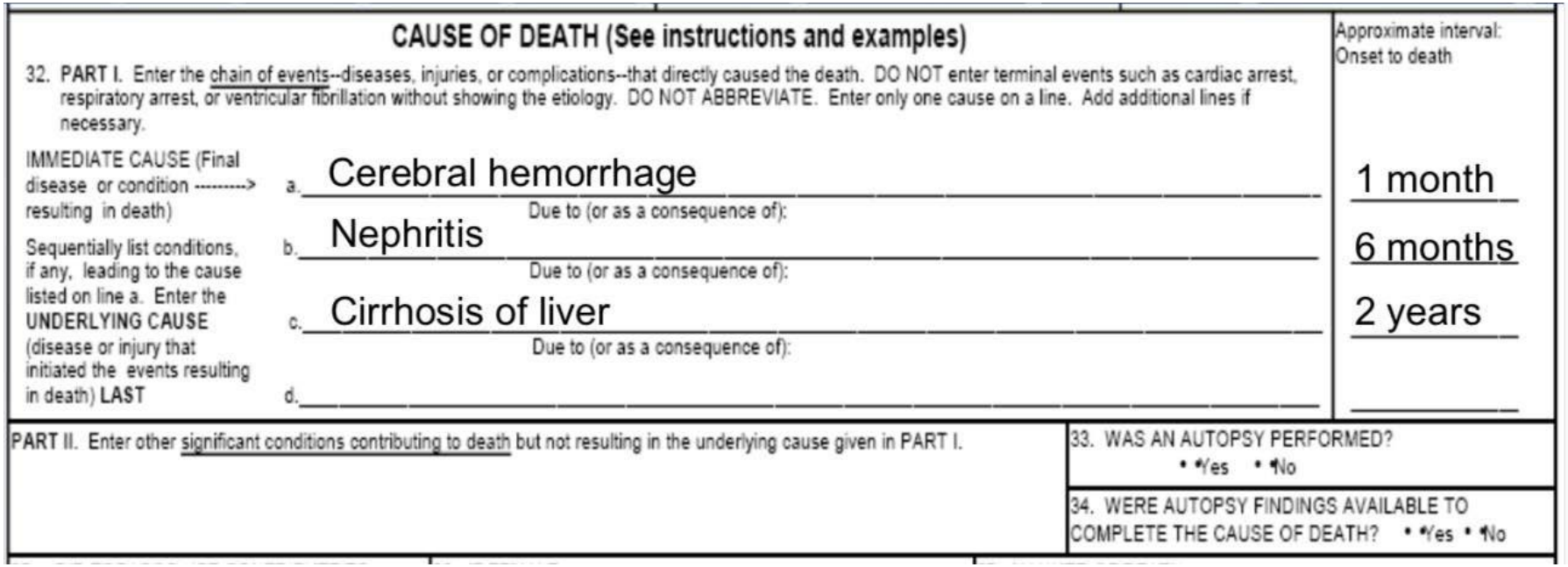
  (Note: pre-Pasteur)
- 1890s—Bertillon (Paris) classification: 161 titles, abstracted to 99, and 44
- 1920 International List of Causes of Death
- 1920s-40s—Manual of the International Statistical Classification of Diseases, Injuries, and Causes of Death
- 1975—ICD-9
- 2015—ICD-10
- ICD-n are under control of the World Health Organization (WHO)

- ICD-9CM, ICD-10CM are US "Clinical Modifications", mainly to support billing

https://www.who.int/classifications/icd/en/HistoryOfICD.pdf

Centers for Disease Control and Prevention (CDC)

- Today, we collect death certificates that record direct & indirect causes of death
- Insurance payments are based on classifications of disease, severity, tests, intervention, …



CAUSE OF DEATH (See instructions and examples)

32. PART I. Enter the chain of events--diseases, injuries, or complications--that directly caused the death. DO NOT enter terminal events such as cardiac arrest, respiratory arrest, or ventricular fibrillation without showing the etiology. DO NOT ABBREVIATE. Enter only one cause on a line. Add additional lines if necessary.

Approximate interval: Onset to death

IMMEDIATE CAUSE (Final disease or condition ---------> resulting in death)	a. Cerebral hemorrhage	1 month
Sequentially list conditions, if any, leading to the cause listed on line a. Enter the UNDERLYING CAUSE (disease or injury that initiated the events resulting in death) LAST	b. Nephritis (Due to (or as a consequence of):)	6 months
	c. Cirrhosis of liver (Due to (or as a consequence of):)	2 years
	d. (Due to (or as a consequence of):)	

PART II. Enter other significant conditions contributing to death but not resulting in the underlying cause given in PART I.

33. WAS AN AUTOPSY PERFORMED? • Yes • No

34. WERE AUTOPSY FINDINGS AVAILABLE TO COMPLETE THE CAUSE OF DEATH? • Yes • No

Courtesy of the CDC. Image is in the public domain.

Quarantine – to prevent epidemics

- Isolation separates sick people with a contagious disease from people who are not sick.
- Quarantine separates and restricts the movement of people who were exposed to a contagious disease to see if they become sick.

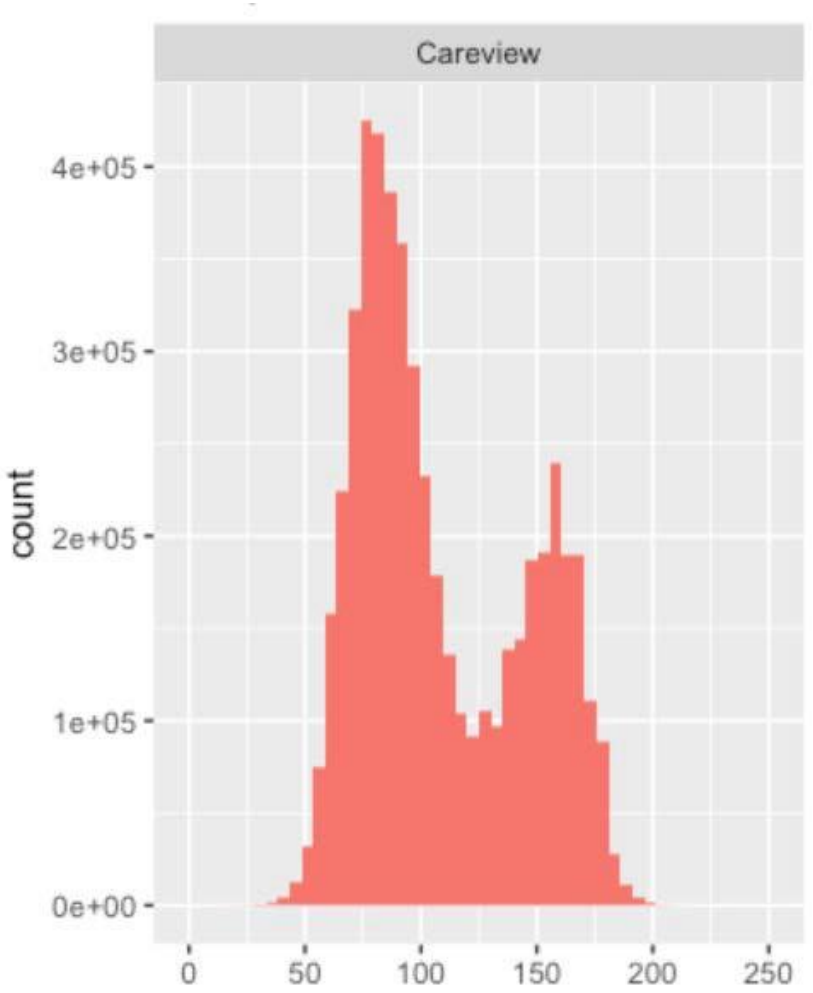
- Mostly used at ports of entry, but sometimes more pervasively
  - Flu
  - Ebola
  - COVID-19

Understanding Clinical Data

- An illustrative example.

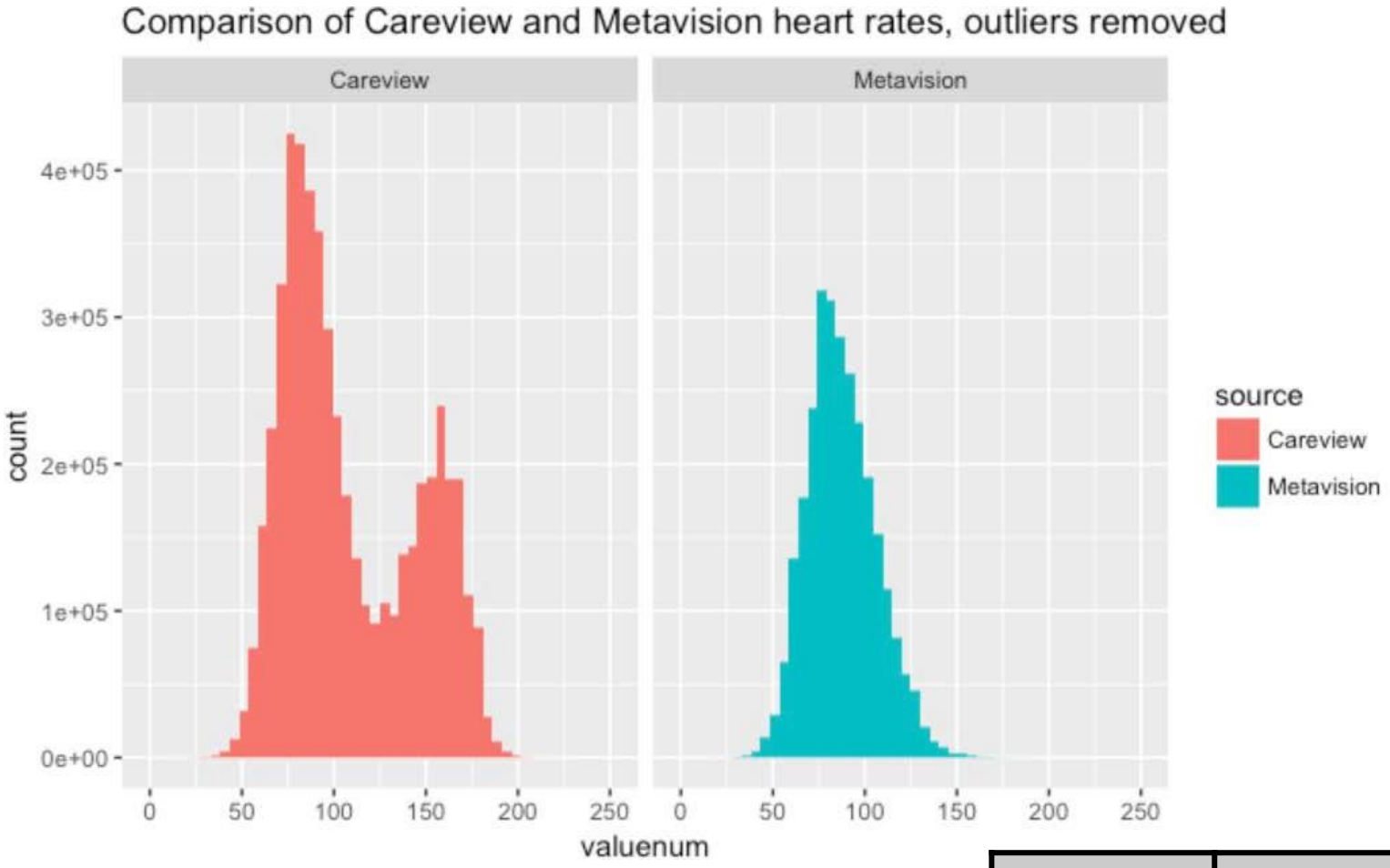
Understanding clinical data

- Consider the distribution of heart rates in the MIMIC-III chart.



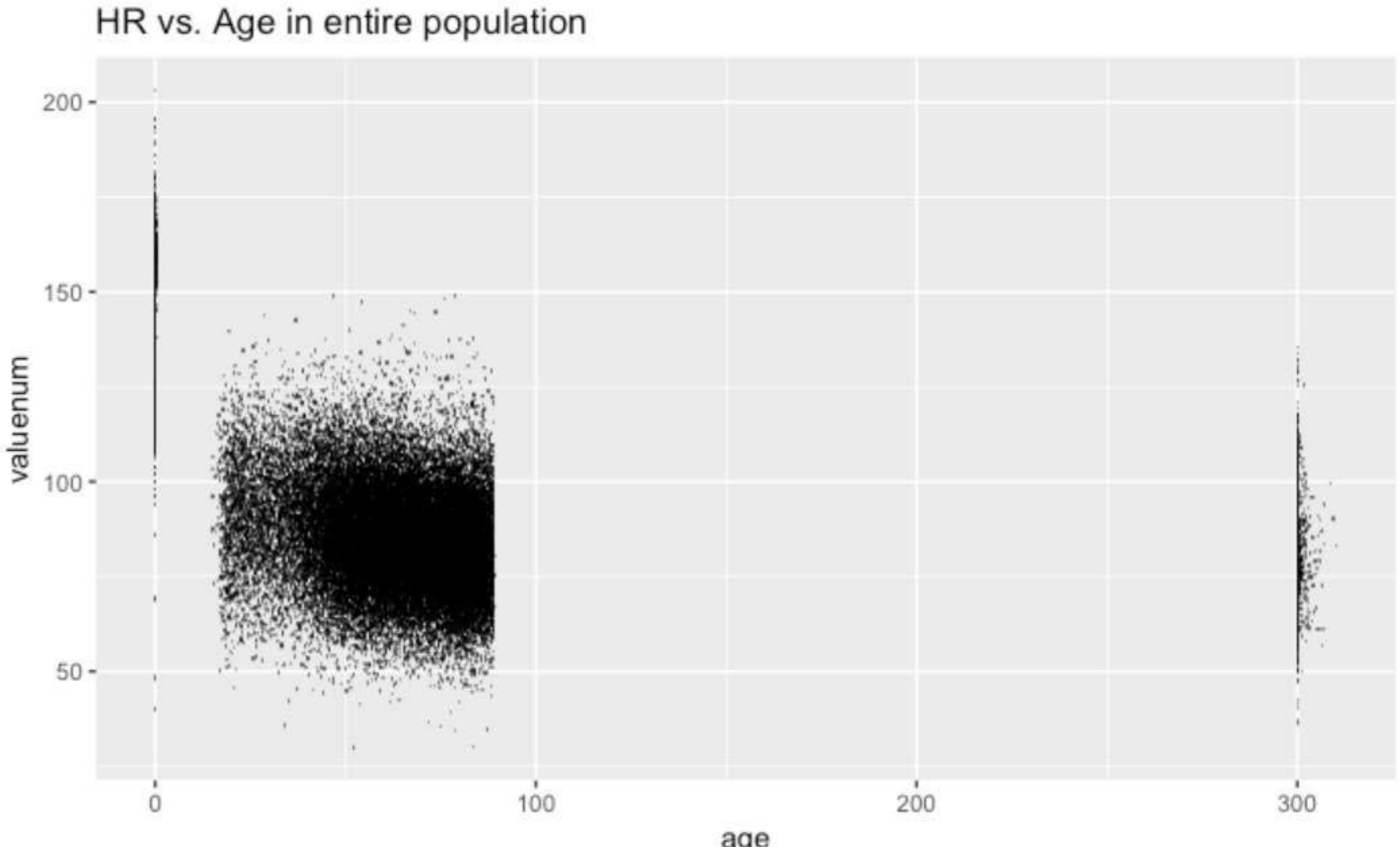
Data from https://mimic.physionet.org

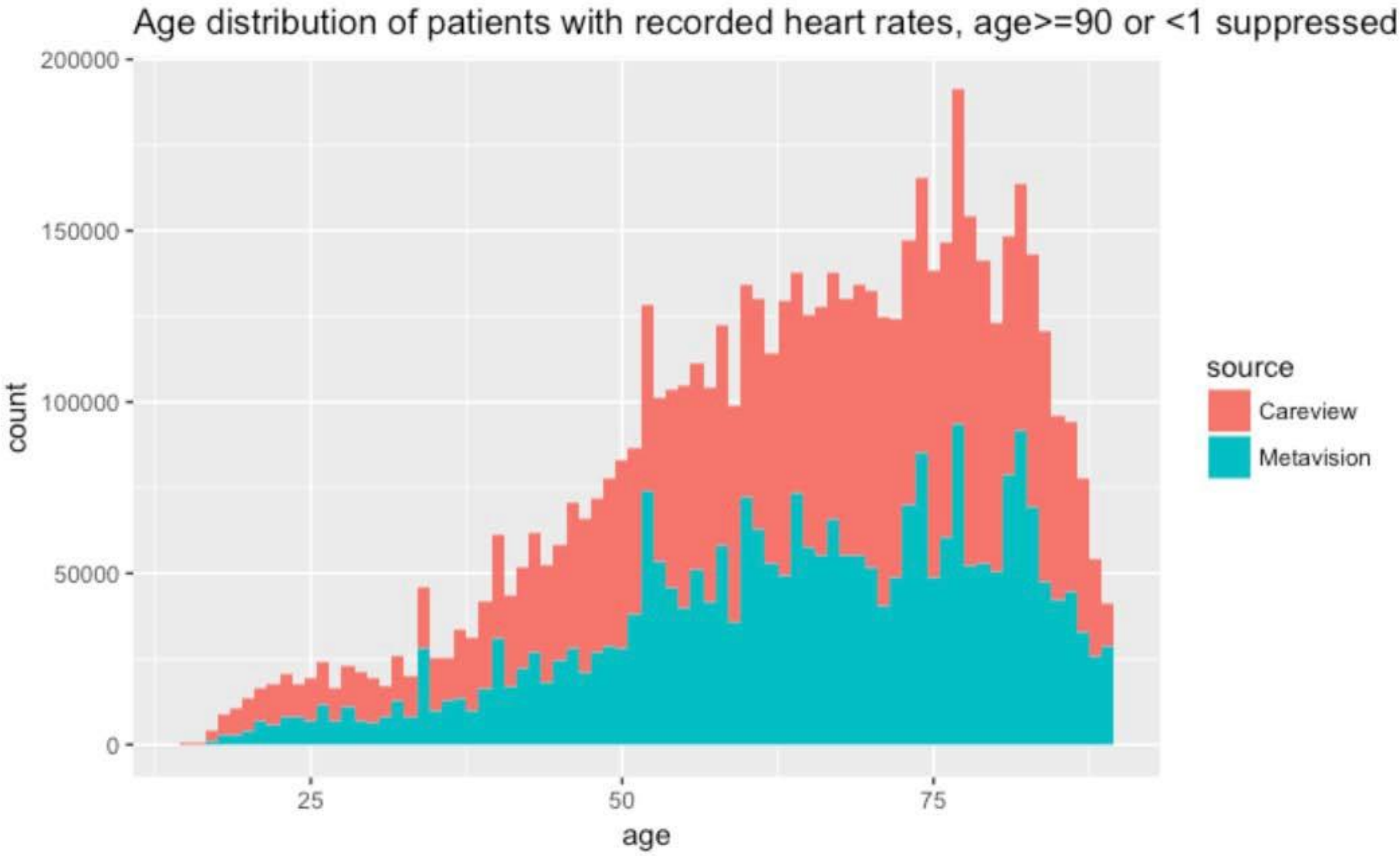
MIMIC-III, a freely accessible critical care database. Johnson AEW, Pollard TJ, Shen L, Lehman L, Feng M, Ghassemi M, Moody B, Szolovits P, Celi LA, and Mark RG. Scientific Data (2016). DOI: 10.1038/sdata.2016.35. Available at: http://www.nature.com/articles/sdata201635

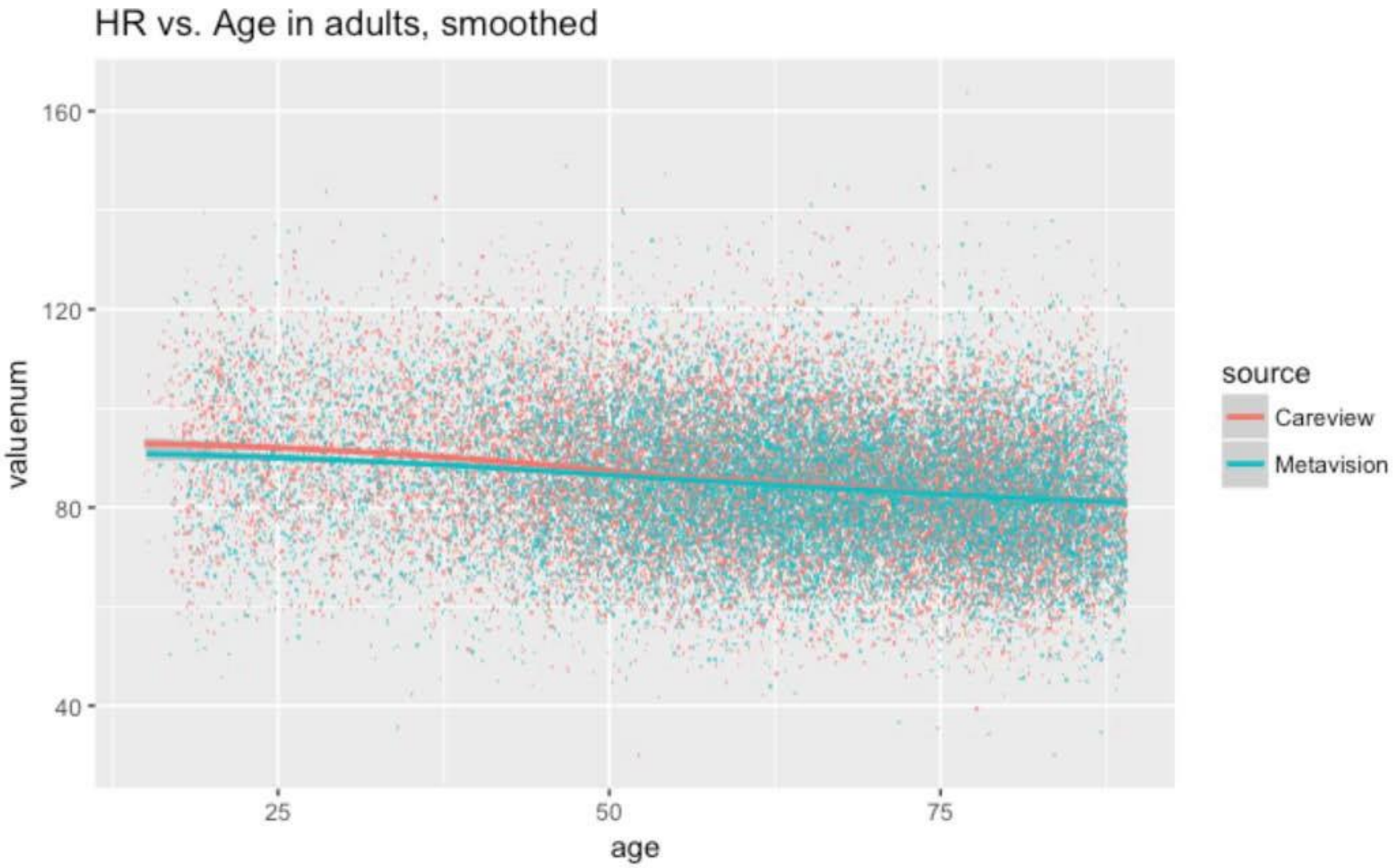


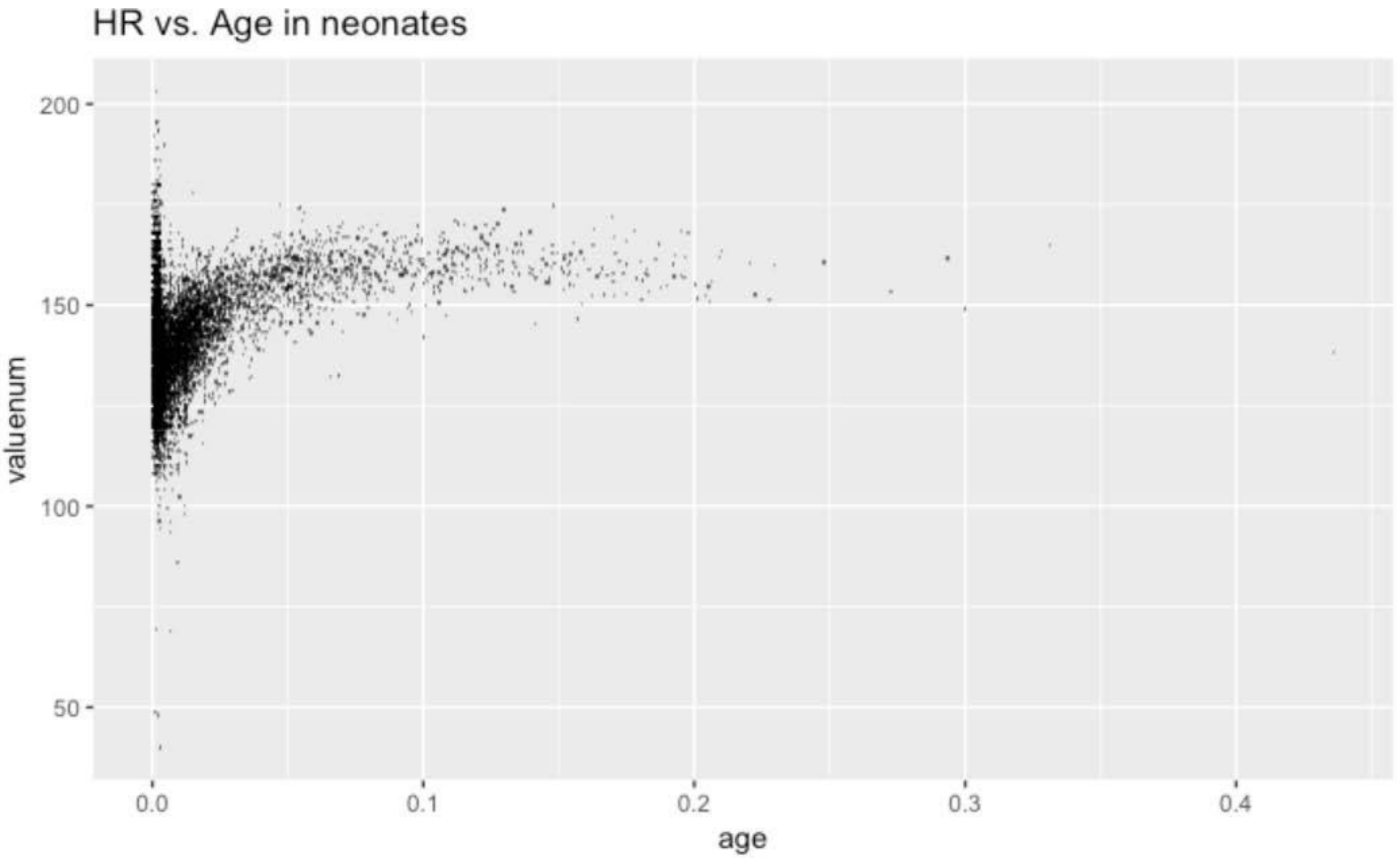
System	n	mean	sd
Careview	5177864	108.89	36.08
Metavision	2761158	87.41	18.72

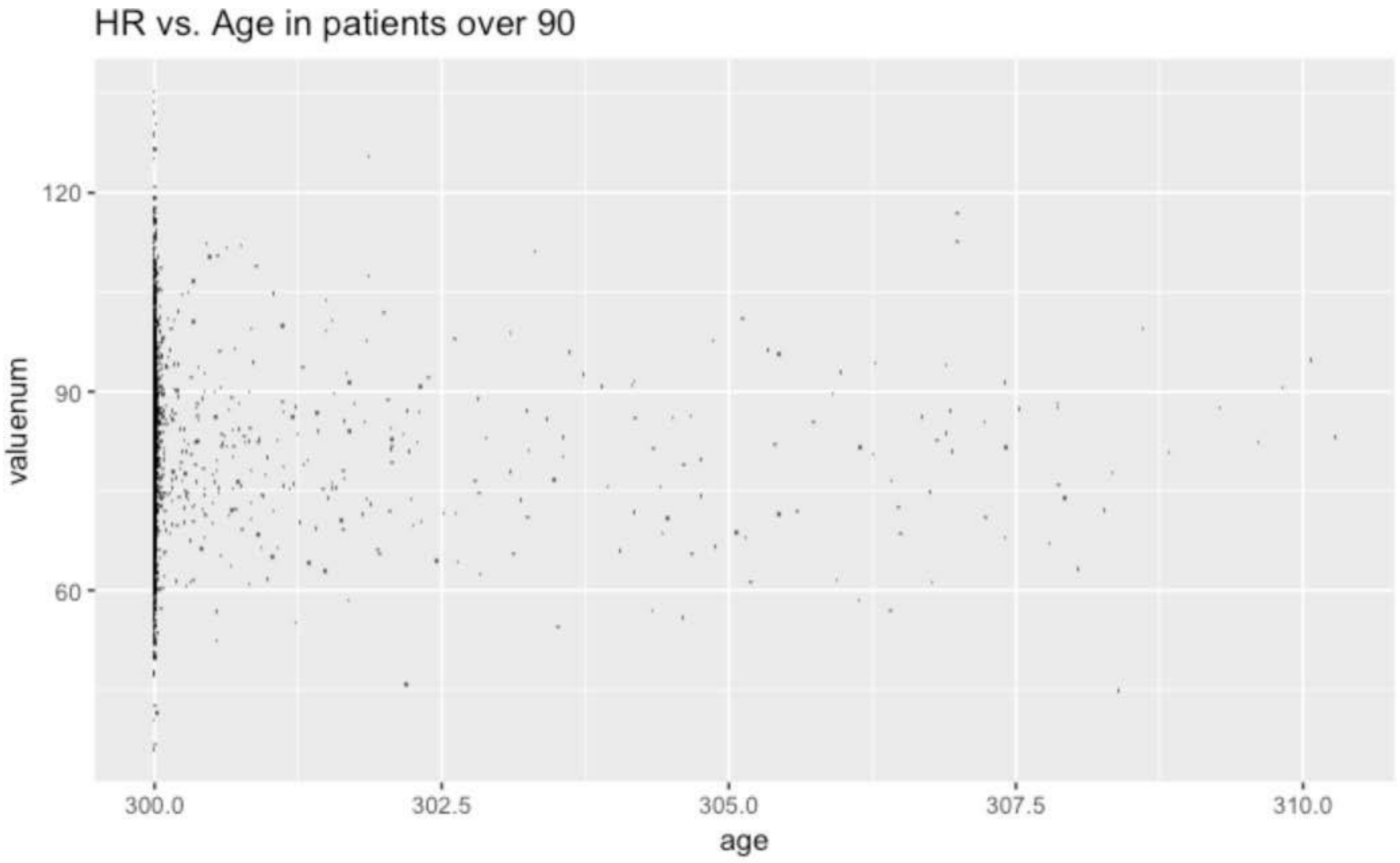
Is Age a Confounder for Heart Rate?




Age distribution of patients with recorded heart rates, age>=90 or <1 suppressed
200000
150000
100000
50000
0
count
25
50
75
age
source
Careview
Metavision


HR vs. Age in adults, smoothed
160
120
80
40
valuenum
25
50
75
age
source
Careview
Metavision


HR vs. Age in neonates
valuenum
200
150
100
50
0.0
0.1
0.2
0.3
0.4
age


HR vs. Age in patients over 90
valuenum
120
90
60
300.0
302.5
305.0
307.5
310.0
age

General Advice

- Visualize!!
  - The data can tell you a lot of stories..
- Clean your data before running models
  - Tons of known and unknown confounders
  - Censored data
  - Missing data
  - Parsing errors
- Always be skeptical!
  - Talk to domain experts.
  - Always get second and third opinion on your results.

Exit pass

https://z.umn.edu/ml4h_exitpass

